



FINAL YEAR PROJECT

Fake News Detection

Using Natural Language Processing & LSTM

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Introduction

1

What is Fake News?

False or misleading information presented as real news.

2

How it Spreads

It can spread quickly through social media and online platforms.

3

Our Goal

This project helps identify whether news is Fake or Real.

FAKE

VS

REAL

Problem Statement

- A large amount of news is available online.
- It is difficult to manually verify every article.
- Fake news can look similar to genuine news.

SOLUTION

An automated system to analyze news quickly.



Objectives

- 1 Detect fake news automatically.
- 2 Analyze the text of a news article.
- 3 Use NLP to prepare the text.
- 4 Train a deep learning model using real and fake news.
- 5 Provide a simple Fake / Real prediction.

How the System Works



A step-by-step pipeline that turns raw news text into a Fake / Real prediction.

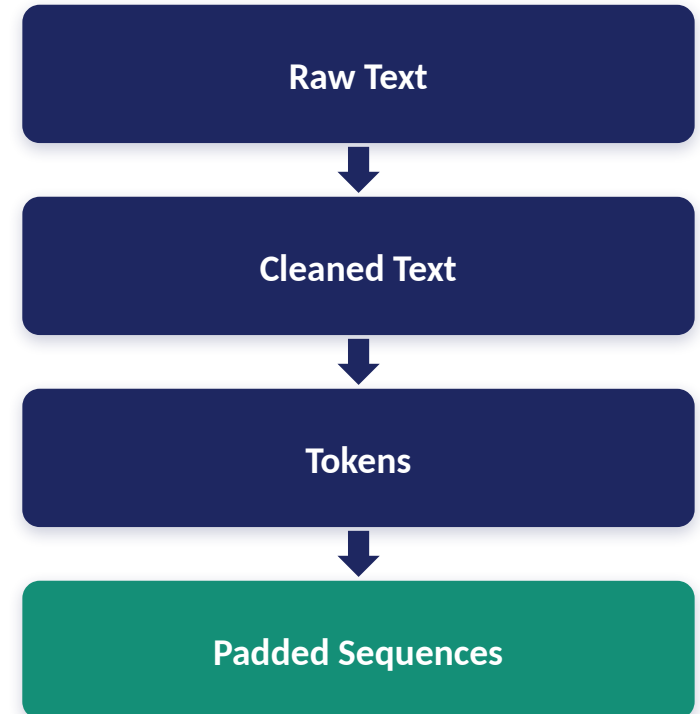
Dataset & Preprocessing

Dataset

Contains labeled Real news
and Fake news articles.

Preprocessing Steps

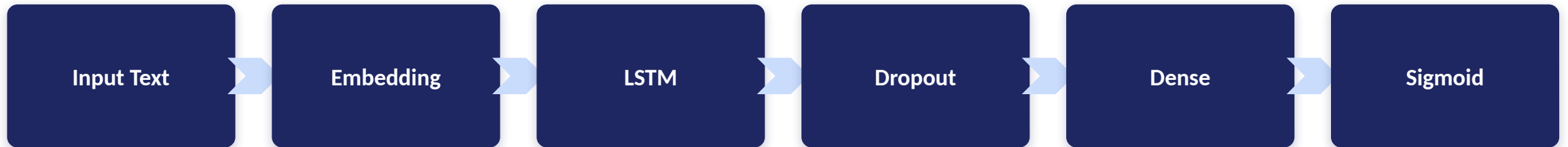
- 1 Clean unnecessary information.
- 2 Tokenize the words.
- 3 Convert words into numerical values.
- 4 Apply padding for model input.



Model Used — LSTM

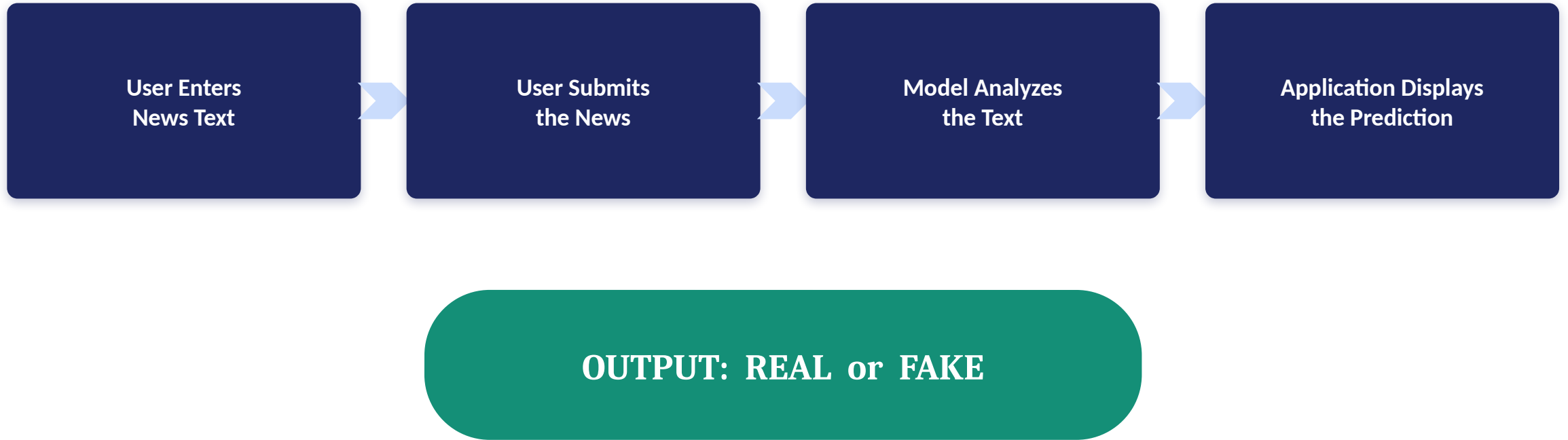
LSTM = Long Short-Term Memory

MODEL FLOW



LSTM is useful for text because it learns relationships between words in a sequence.

Application



Results & Benefits

- 1 The model classifies news using patterns learned during training.
- 2 Predictions are provided quickly.
- 3 Simple and easy-to-use interface.
- 4 Helps users identify potentially misleading news.
- 5 Useful as an initial screening tool.

Conclusion

- Developed a Fake News Detection application.
- Used NLP to process news text.
- Used an LSTM model for classification.
- The system predicts whether news is Fake or Real.
- Demonstrates a practical use of AI, NLP, and deep learning.

Thank You

Questions?